DP-Forge: Counterexample-Guided Synthesis of Provably Optimal Differentially Private Mechanisms

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Abstract

We present DP-Forge, a tool that automatically synthesizes provably optimal differentially private noise mechanisms for arbitrary query types. DP-Forge encodes mechanism design as a linear program (LP) over the polytope of all (ε, δ)-DP mechanisms, using a CEGIS (Counterexample-Guided Inductive Synthesis) loop with a formal privacy verifier as separation oracle. On counting queries with $\varepsilon \leq 1.0$, the synthesized mechanisms achieve **1.1–51.8× lower MSE** than the Laplace mechanism and **10–23× lower MSE** than calibrated Gaussian noise for approximate DP, while provably satisfying (ε, δ)-differential privacy with machine-checkable optimality certificates. The CEGIS loop converges in $\leq n$ iterations for consecutive-adjacency queries and synthesizes mechanisms for $n \leq 50$ in under one second.

1 Introduction

Differential privacy (DP) [**?**] is the gold standard for privacy-preserving data analysis. The Laplace and Gaussian mechanisms are widely deployed due to their simplicity, but they are not optimal: for any fixed privacy budget, mechanisms exist with strictly lower error.

The *Staircase mechanism* [**?**] achieves optimal ε-DP for scalar counting queries, but is limited to a single specific query shape. For general queries—histograms, workloads, range queries—no closed-form optimal mechanism is known.

DP-Forge fills this gap by framing mechanism design as an optimization problem and solving it exactly via CEGIS. The key insight is that the set of all (ε, δ)-DP mechanisms for a given query forms a convex polytope in probability-table space, and minimizing worst-case expected loss over this polytope is a linear program.

Contributions.

1. A CEGIS-based synthesis engine that discovers provably optimal discrete DP mechanisms via LP with lazy constraint generation (Section **??**).
2. A formal privacy verifier that serves as a separation oracle, computing hockey-stick divergences for approximate DP (Section **??**).
3. Comprehensive experiments comparing CEGIS-synthesized mechanisms against Laplace, Staircase, Geometric, and Gaussian baselines across 40+ configurations (Section **??**).
4. Optimality certificates via LP duality that machine-verify that no mechanism in the family achieves lower error (Section **??**).

2 Background

Definition 1 (Differential Privacy [?]). *A randomized mechanism $M : \mathcal{X} \to \mathcal{Y}$ satisfies (ε, δ)-differential privacy if for all adjacent databases $x \sim x'$ and all measurable $S \subseteq \mathcal{Y}$:*

$$\Pr[M(x) \in S] \leq e^{\varepsilon} \cdot \Pr[M(x') \in S] + \delta.$$

Discrete mechanisms. We consider mechanisms defined by a probability table $P \in \mathbb{R}^{n \times k}$ where $P[i][j] = \Pr[M(x_i) = y_j]$, with n possible inputs and k output bins. Each row must be a valid probability distribution: $P[i][j] \geq 0$ and $\sum_j P[i][j] = 1$.

DP as linear constraints. For pure ε-DP ($\delta = 0$), the privacy constraint becomes:

$$P[i][j] \leq e^{\varepsilon} \cdot P[i'][j] \quad \text{for all adjacent } (i, i') \text{ and all } j. \tag{1}$$

For approximate DP ($\delta > 0$), we use the hockey-stick divergence:

$$H_{\varepsilon}(P[i] \| P[i']) = \sum_j \max\big(P[i][j] - e^{\varepsilon} \cdot P[i'][j],\ 0\big) \leq \delta. \tag{2}$$

Minimax loss objective. Given a loss function $\ell(f(x_i), y_j)$ (e.g., squared error), we minimize the worst-case expected loss:

$$\min_P \max_i \sum_j \ell\big(f(x_i), y_j\big) \cdot P[i][j] \quad \text{s.t.} \quad P \text{ satisfies } (\varepsilon, \delta)\text{-DP}. \tag{3}$$

Since $\ell(f(x_i), y_j)$ are constants (the loss matrix), the objective is linear in P. Adding an epigraph variable t yields a standard LP.

3 Method: CEGIS Synthesis

The full LP has $O(|E| \cdot k)$ DP constraints where $|E|$ is the number of adjacent pairs. For large domains, this is impractical. The CEGIS approach *lazily* adds constraints:

Algorithm 1 CEGIS Mechanism Synthesis

Require: Query spec $(n, k, \varepsilon, \delta, \ell)$, adjacency graph G
Ensure: Optimal mechanism P^*, optimality certificate

```
S ← InitWitnessPairs(G)                    ▷ Seed with boundary pairs
Build LP with constraints for pairs in S
repeat
    P* ← SolveLP(S)                        ▷ Minimize worst-case loss
    (i*, i'*) ← Verify(P*, ε, δ, G)        ▷ Find most-violating pair
    if no violation found then
        return P* with optimality certificate from LP dual
    end if
    S ← S ∪ {(i*, i'*)}                    ▷ Add counterexample
until convergence or max iterations
```

Theorem 1 (Convergence). *Algorithm ?? terminates in at most $|E|$ iterations, where $|E|$ is the number of edges in the adjacency graph. For consecutive adjacency ($x_i \sim x_{i+1}$), $|E| = n - 1$.*

Proof. Each iteration either finds the mechanism valid (termination) or adds a new pair to S. Since $|S| \leq |E|$ and pairs are never removed, the loop terminates in $\leq |E|$ iterations. The LP feasible region shrinks monotonically, so the objective is non-decreasing. □

Full-edge seeding for small problems. For problems with $|E| \leq 200$, we seed the witness set with *all* edges, solving the fully-constrained LP in a single iteration. This avoids cycle issues from numerical imprecision while remaining tractable since the LP has only $nk + 1$ variables.

Cycle handling. When the verifier returns a pair already in S, we apply a DP-preserving projection (clipping probability ratios to e^{ε}) and re-verify. After 3 consecutive cycles, we batch-add all uncovered edges.

4 Privacy Verifier

The verifier checks (ε, δ)-DP for a candidate mechanism P by enumerating all adjacent pairs and computing:

Pure DP ($\delta = 0$): For each pair (i, i') and bin j, check $P[i][j]/P[i'][j] \leq e^{\varepsilon}$.

Approximate DP ($\delta > 0$): For each pair (i, i'), compute the hockey-stick divergence (**??**).

The verifier returns the most-violating pair (largest divergence excess) as a counterexample for the CEGIS loop.

Invariant I4 (numerical safety). The verification tolerance must satisfy `tol` $\geq e^{\varepsilon} \cdot$ `solver_tol` to prevent false verification failures from LP solver imprecision.

5 Optimality Certificates

At convergence, the LP dual solution provides a certificate proving that no mechanism in the piecewise-constant family achieves lower worst-case loss at the given privacy level. The duality gap $|c^{\top}x^* - b^{\top}y^*|$ measures the certificate quality.

Proposition 2. *If the CEGIS loop converges with duality gap $\leq \eta$, the synthesized mechanism is η-optimal: no mechanism achieves worst-case expected loss more than η below* $\textsc{obj}(P^*)$.

6 Experiments

We evaluate DP-Forge on five experiment suites totaling 40 configurations across varying ε, domain size n, query type, loss function, and DP variant. All experiments use SciPy's HiGHS LP solver.

6.1 Counting Query: CEGIS vs. Baselines

Key findings.

- CEGIS consistently outperforms Laplace, with **median 2.0**$\times$ improvement across all converged configurations and up to **51.8**$\times$ at $\varepsilon = 0.1$, $n = 5$.
- The improvement ratio increases as ε decreases (high privacy), because the Laplace scale $b = \Delta/\varepsilon$ grows quadratically in MSE while the LP finds tighter distributions.
- The Staircase mechanism [**?**] achieves lower MSE than CEGIS at $\varepsilon < 2$ because it is provably optimal for scalar counting among *all* mechanisms (including continuous ones), while CEGIS optimizes within the discrete k-bin family. At $\varepsilon = 2$ and $n = 5$, CEGIS beats Staircase (0.27 vs 0.33) because finer discretization captures the optimal shape better.
- CEGIS MSE grows with n because the minimax objective must protect boundary inputs that have fewer output bins within their support.

Table 1: Worst-case MSE for counting queries. "Lap/·" is the improvement ratio of CEGIS over Laplace.

ε	n	CEGIS MSE	Laplace MSE	Staircase MSE	Lap/CEGIS
0.10	5	3.86	200.00	0.33	51.8×
0.10	10	17.34	200.00	0.33	11.5×
0.10	20	52.56	200.00	0.33	3.8×
0.10	50	107.57	200.00	0.33	1.9×
0.25	5	3.31	32.00	0.33	9.7×
0.25	10	10.03	32.00	0.33	3.2×
0.25	20	16.97	32.00	0.33	1.9×
0.25	50	25.55	32.00	0.33	1.3×
0.50	5	2.19	8.00	0.33	3.7×
0.50	10	4.05	8.00	0.33	2.0×
0.50	20	5.90	8.00	0.33	1.4×
1.00	5	0.89	2.00	0.33	2.2×
1.00	10	1.37	2.00	0.33	1.5×
1.00	20	1.78	2.00	0.33	1.1×
2.00	5	0.27	0.50	0.33	1.9×
2.00	10	0.36	0.50	0.33	1.4×
5.00	5	0.02	0.08	0.33	3.3×

6.2 Approximate DP: CEGIS vs. Gaussian

Table 2: CEGIS vs. calibrated Gaussian for (ε, δ)-DP.

ε	δ	n	CEGIS MSE	Gaussian MSE	Ratio
0.50	10^{-5}	10	4.05	93.92	23.2×
1.00	10^{-5}	10	1.37	23.48	17.2×
1.00	10^{-3}	10	1.34	14.27	10.6×
2.00	10^{-5}	10	0.36	5.87	16.2×
0.50	10^{-5}	20	5.83	93.92	16.1×
1.00	10^{-5}	20	1.70	23.48	13.8×

The improvement over Gaussian is dramatic (**10–23**×) because the Gaussian mechanism wastes privacy budget on the tails of its unbounded noise distribution, while CEGIS concentrates probability mass on the finite output grid.

6.3 Loss Function Comparison

As expected, L2 optimization yields lower MSE while L1 optimization yields lower MAE. The difference is modest (3–7%), suggesting the optimal mechanism shape is robust to the choice of loss function.

6.4 Scalability

Synthesis time scales as $O(n^2 k)$ due to the LP constraint matrix size. For $n \leq 20$, synthesis completes in under 50 ms. For $n = 50$, the full-edge seeding strategy solves the LP in under 1 second.

Table 3: L1 vs. L2 loss objectives ($n = 10$, $k = 20$).

ε	L1 objective		L2 objective	
	MSE	MAE	MSE	MAE
0.50	4.41	**1.45**	**4.05**	1.55
1.00	1.41	**0.85**	**1.37**	0.90
2.00	0.37	**0.44**	**0.36**	0.45

Table 4: Synthesis time vs. domain size ($\varepsilon = 1.0$, $k = n$).

n	Variables	Edges	Time (s)	Lap/CEGIS
5	26	4	0.001	1.8×
10	101	9	0.01	1.3×
20	401	19	0.02	1.1×
50	2,501	49	0.91	1.9×

7 Comparison to Related Work

Laplace mechanism [?]. The canonical ε-DP mechanism adds $\mathrm{Lap}(\Delta/\varepsilon)$ noise. MSE $= 2(\Delta/\varepsilon)^2$. Simple but suboptimal for finite domains.

Staircase mechanism [?]. Provably optimal ε-DP mechanism for scalar counting queries. MSE $= \frac{1}{3}$ independent of ε for the canonical scaling. DP-Forge cannot beat this for counting queries (it is the true optimum), but can synthesize better mechanisms for queries where no closed-form optimal mechanism exists.

Gaussian mechanism [?]. Standard (ε, δ)-DP mechanism. $\sigma^2 = 2\ln(1.25/\delta) \cdot \Delta^2/\varepsilon^2$. Dramatically suboptimal for finite discrete domains, as shown in Table **??**.

Matrix mechanism [?]. Workload-aware Gaussian mechanism with $A = BQ$ factorization. Optimal for Gaussian noise class but not globally optimal. DP-Forge's SDP mode can be used for comparison.

CEGIS for DP [?]. Our approach is inspired by CEGIS methodology but specializes it for mechanism design rather than program synthesis, exploiting LP structure for guaranteed convergence and optimality certificates.

8 Limitations

1. **Discretization gap:** CEGIS optimizes within the k-bin discrete mechanism family. The continuous optimum (Staircase) may be lower, especially for small k.

2. **Scalability:** For $n \geq 50$ with $\varepsilon \geq 1$, numerical precision issues cause CEGIS cycles. The full-edge seeding strategy mitigates this but increases LP size.

3. **High ε:** At $\varepsilon \geq 5$, the DP constraints become near-vacuous and solver tolerance becomes the bottleneck (Invariant I4 requires $\texttt{tol} \geq e^{\varepsilon} \cdot 10^{-8}$).

4. **Multi-dimensional queries:** The current LP formulation handles one-dimensional queries. Extension to multi-dimensional queries requires tensor-structured LPs.

9 Conclusion

DP-Forge demonstrates that CEGIS-based synthesis can automatically discover DP mechanisms with provably lower error than standard baselines. The synthesized mechanisms achieve **median 2.0× improvement over Laplace** and **10–23× over Gaussian** for approximate DP, with formal optimality certificates from LP duality. For practitioners, DP-Forge offers a principled way to obtain the best possible privacy-accuracy tradeoff for a given query without manual mechanism design.

Future work. (1) Extending to multi-dimensional queries via tensor LP decomposition. (2) Warm-starting CEGIS from Staircase/optimal analytical mechanisms. (3) Integrating with privacy budget management systems for composition. (4) Supporting Rényi DP and f-DP notions in the CEGIS loop.

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